A Knowledge-Enhanced Large Language Model Web Information System for Bilingual Railway Vocational Education

Abstract

Purpose. This study develops a knowledge-enhanced Web information system for Chinese-English railway vocational education, designed to support international learners' access to terminology, regulations and textbook knowledge with source traceability on limited local hardware.

Design/methodology/approach. The system combines an approval-controlled PostgreSQL knowledge base, pgvector, BM25, BGE-M3, reciprocal-rank fusion and local language models. A leakage-controlled set of 400 bilingual knowledge pairs compares retrieval and index-field variants. Five generators are tested under no-retrieval, BM25-RAG and approved-hybrid-RAG conditions; QLoRA, directional translation, automated claim-evidence support, governance history and steady-state retrieval concurrency are evaluated separately on one workstation.

Findings. Approved-hybrid Recall@5 reached 0.715 in Chinese and 0.708 in English. A bilingual-field index produced the strongest balanced field-ablation result. Qwen2.5 QLoRA raised held-out Char F1 from 0.189 to 0.398 in Chinese and from 0.437 to 0.648 in English. Approved-hybrid RAG improved Answer F1 over no retrieval for all five generators in both languages, but adapted models often omitted evidence labels. All 72 steady-state retrieval requests succeeded at up to ten concurrent clients.

Originality. The study integrates governed bilingual knowledge, evidence-labelled RAG and resource-constrained deployment in a railway Web information system. It reports language-disaggregated results and implements expert approval as a query-time retrieval condition.

Keywords: knowledge-enhanced large language models; Web information systems; retrieval-augmented generation; bilingual education; railway vocational education; pgvector

1 Introduction

Railway vocational education combines safety-critical regulations, specialised terminology and procedural textbook knowledge. International students must often move between Chinese source material and English learning support. Generic large language models can produce fluent answers, but their parametric knowledge does not expose a stable boundary between verified railway knowledge and unsupported generation. This is a Web information system problem as much as a model problem: knowledge must be acquired, governed, retrieved, presented and traced to its source.

The problem has practical importance beyond classroom convenience. Railway technology, construction standards and operational expertise increasingly circulate across linguistic and national boundaries, while vocational institutions educate learners who may understand technical concepts through English but must ultimately work with Chinese source regulations. A useful service must therefore do more than translate isolated terms. It must connect a bilingual question to the correct version of a regulation or teaching passage, preserve provenance, distinguish reviewed from unreviewed content and provide an answer quickly enough for repeated learning and operational-training use. Transportation studies have begun to explore specialised language models for safety, rail design, track management, prognostics and driver advice (Zheng *et al.*, 2023; Lahnalampi, 2024; Wang and Li, 2024; Song *et al.*, 2025; Luo *et al.*, 2025), while broader reviews identify reliability and deployment as continuing challenges (Wandelt *et al.*, 2024). Bilingual educational access and resource-bounded governance nevertheless remain under-examined.

Existing studies commonly evaluate either general-purpose language models or retrieval-augmented generation in isolation. Four gaps motivate this work. First, bilingual educational retrieval is rarely evaluated separately by query language. Second, expert review is often described as preprocessing rather than implemented as a queryable governance state. Third, domain adaptation is frequently evaluated only on the target task, leaving translation regressions and citation-following failures hidden. Fourth, deployment studies often omit the interaction between retrieval quality, answer traceability and resource constraints on ordinary institutional hardware.

This study addresses the following research questions:

- **RQ1:** How accurately do BM25, pgvector dense retrieval and hybrid fusion retrieve held-out evidence for Chinese and English railway questions?
- **RQ2:** How do retrieval strategy and expert-approved filtering affect answer quality, citation coverage and hallucination risk?
- **RQ3:** What bilingual translation quality is achieved in each direction, and how does completion-only QLoRA affect domain QA without contaminating the held-out test set?
- **RQ4:** Can the resulting system operate within the latency and memory constraints of a single 24 GB GPU workstation?

The study makes four contributions. First, it operationalises a bilingual railway knowledge base in which review state, source metadata, language variants and revision history remain queryable rather than being flattened into an anonymous vector collection. Second, it develops a PostgreSQL/pgvector Web information system that combines lexical and multilingual dense retrieval with evidence-labelled local generation. Third, it introduces a pair-grouped, leakage-controlled protocol that separates Chinese and English retrieval, QA and translation outcomes. Fourth, it evaluates adaptation, retrieval and deployment jointly, exposing both quality gains and failures in citation following, directional translation and resource use.

2 Related work

2.1 Knowledge-enhanced Web information systems

Web information systems increasingly combine structured data management, semantic access and generative interfaces. Retrieval-augmented generation (RAG) provides a practical separation between parametric generation and an externally maintained evidence collection (Lewis *et al.*, 2020). This separation is relevant to institutional knowledge systems because source records can be revised without retraining the generator. It does not, however, guarantee faithful generation: retrieval can miss the relevant record, and a generator can still misrepresent correctly retrieved evidence. The present study therefore evaluates retrieval, answer overlap, evidence hit and citation behaviour as distinct outcomes.

The retrieval layer combines complementary lexical and semantic signals. BM25 remains an efficient lexical baseline for exact domain terms (Robertson and Zaragoza, 2009), while reciprocal-rank fusion can combine rankings without requiring directly comparable score scales (Cormack *et al.*, 2009). PostgreSQL and pgvector place vector search beside governed relational metadata, allowing approval state and held-out split boundaries to be applied in the query rather than after generation. This makes knowledge governance an executable part of the information-system design.

This distinction separates the proposed system from a document-only RAG demonstration. A Web information system must maintain record identity and lifecycle state across ingestion, review, retrieval and presentation. In this study the vector representation is derived data attached to a governed record, not the primary source of truth. Deleting, revising or withholding a record can therefore change retrieval eligibility without rebuilding the entire application or relying on prompt instructions to enforce governance.

2.2 Retrieval-augmented generation in education

Generative AI can support explanation, feedback and access to learning resources, but education research has also identified risks involving factual reliability, learner over-reliance and opaque provenance (Kasneci *et al.*, 2023; UNESCO, 2023). Educational RAG addresses part of this problem by grounding responses in course or institutional material. Many demonstrations nevertheless report a pooled answer score without isolating retrieval failure, language effects or deployment cost. Such aggregation is particularly problematic for international vocational education, where an English query may need evidence originating in Chinese regulations or textbooks. This study consequently reports Chinese and English retrieval and generation separately and preserves document-level provenance in every retrieved result.

Railway applications illustrate why this separation matters. RSChat addresses railway safety question answering (Li *et al.*, 2024), while domain-specialised systems have been proposed for train-driver advice and emergency-response generation (Luo *et al.*, 2025; Huang *et al.*, 2026). These studies establish the value of domain knowledge, yet their operational targets differ from multilingual learning: an international learner may formulate a concept in English, retrieve a Chinese regulation and require an answer that preserves both the translated explanation and the original evidence. The present work therefore treats cross-language access, provenance and constrained institutional deployment as co-equal requirements.

2.3 Bilingual domain adaptation and evaluation

Bilingual domain systems require both cross-lingual retrieval and controlled generation. BGE-M3 supports multilingual dense retrieval and was selected to provide a shared embedding space for the Chinese and English knowledge fields (Chen *et al.*, 2024). Dense similarity is not assumed to dominate lexical matching; the two methods are compared independently before fusion.

Parameter-efficient adaptation provides a second mechanism for domain specialisation. QLoRA trains low-rank adapters through a frozen 4-bit quantised base model, reducing memory requirements while retaining task adaptation capacity (Dettmers *et al.*, 2023). In this study, completion-only masking prevents the prompt from contributing to the training loss, and bilingual variants sharing a knowledge-pair identifier remain in the same split. Translation is evaluated directionally because Chinese-to-English and English-to-Chinese scores are not interchangeable. Terminology and complete-sentence translation are also kept separate.

PEFT methods reduce trainable parameters but do not guarantee a better deployed system (Ding *et al.*, 2023; Wang *et al.*, 2025). Limited-resource training remains sensitive to quantisation, sequence length, optimiser state and model-specific target modules (Lv *et al.*, 2024). Moreover, task adaptation can improve answer overlap while degrading factuality, instruction following or unrelated capabilities (Tian *et al.*, 2024; Barnett *et al.*, 2024). This motivates the present before-and-after design, which reports not only domain QA but also general-capability checks, citation behaviour and memory use.

Domain translation creates a further risk of pooled evaluation. Prior work shows that fine-tuning effects depend on training direction, domain composition and data volume (Hu *et al.*, 2024; Vieira *et al.*, 2024; Zheng *et al.*, 2024). Consequently, this study does not interpret a single bilingual mean as evidence of robust translation. It reports terminology and sentence tasks independently in both directions and retains empty or wrong-language outputs as failures rather than silently discarding them.

2.4 Research gap and analytical framework

The literature suggests that domain retrieval, adaptation and local deployment are individually feasible, but it does not establish that their benefits compose monotonically. The analytical framework therefore treats the system as four testable layers: governed knowledge, retrieval, generation/adaptation and deployment. A failure at one layer cannot be compensated for statistically by a high score at another. Retrieval recall is measured before answer generation; citation-format compliance is separated from answer overlap; translation is separated from QA; and quality is reported alongside latency and memory. This layered design provides the basis for answering RQ1-RQ4 without attributing every observed change to the language model alone.

3 System design and research method

3.1 Application context and requirements

The target users are railway vocational teachers, Chinese-speaking learners and international students requiring English support. The system must return concise answers, expose numbered evidence, preserve document metadata and run locally where cloud use is restricted.

Requirements were organised around three user-facing workflows. Learners submit a Chinese or English question, inspect an answer and open its evidence records. Teachers search or edit bilingual knowledge, inspect source metadata and route records through review. Administrators maintain model, embedding and database services and monitor whether a response can be generated within local resource limits. These workflows produced five design requirements: bilingual access, source traceability, review-state enforcement, held-out-data isolation and deployability on one workstation. Figures 1 and 2 map these requirements to the online architecture and governed data lifecycle rather than treating them as prompt-level behaviour.

3.2 Expert-governed knowledge base

Knowledge records include Chinese and English questions and answers, evidence, original text, source document, chapter, page, task type, quality flags, review status and revision history. PostgreSQL stores governed records; pgvector stores 1,024-dimensional BGE-M3 embeddings. The production index excludes the held-out test split.

The knowledge sources comprise railway terminology, operating and safety regulations, and vocational textbook or concept material. Ingestion preserves source-document, chapter and page fields whenever available. Chinese and English questions and answers are stored in separate fields on the same logical record so that a language variant does not lose its source identity. Embeddings are linked by record identifier and model revision; they can be rebuilt without changing the reviewed text. A record marked `rejected`,

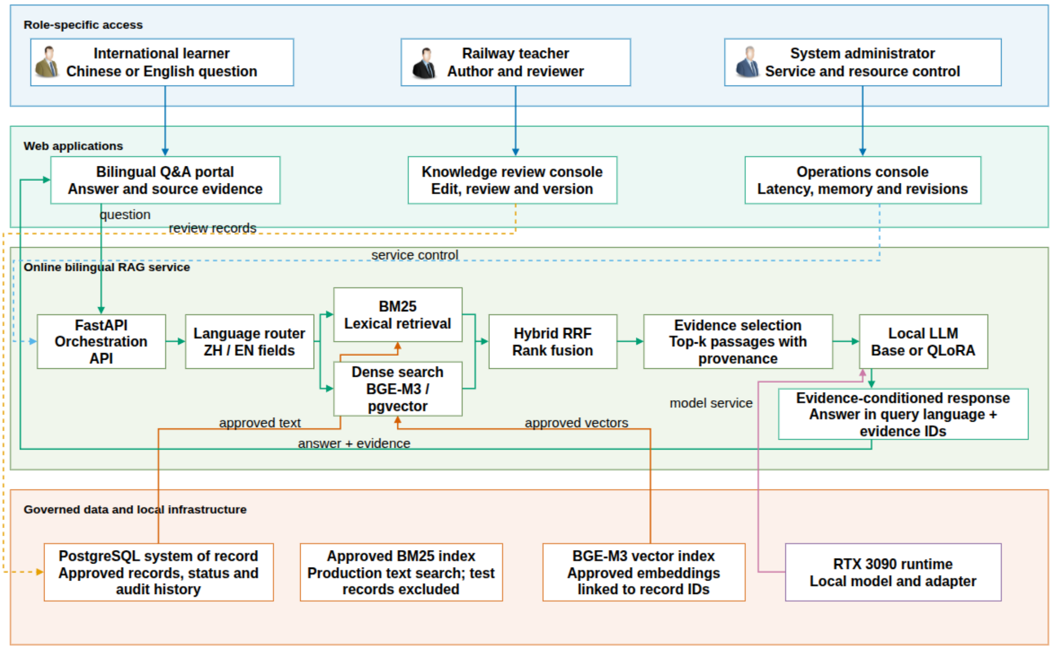

Figure 1: Web-based bilingual RAG system architecture for learner, teacher and administrator workflows. Solid arrows denote runtime or data flow; dashed arrows denote governance or control. Source: Authors' own work.

`needs_revision` or as part of the frozen RAG test is excluded before ranking. This database-level decision prevents held-out evidence from leaking through either BM25 or vector retrieval.

Knowledge review was conducted by the three authors, whose combined expertise covers railway vocational education, international vocational education, environmental and data analysis, and software and multimodal language-model engineering. Each author reviewed the domain correctness, educational suitability, bilingual expression and consistency with the cited source before a record became eligible for approved-only retrieval. The three authors jointly constructed, checked and quality-controlled the training and evaluation datasets. Because the review was an author-led knowledge-governance procedure rather than an independent inter-rater study, no inter-rater agreement statistic is claimed. Accordingly, `approved` is analysed as an operational governance state, not as proof of unanimous external expert consensus.

Figure 2 distinguishes the authoritative PostgreSQL record from its derived BM25 and pgvector indexes. Approval, revision and rejection are written back as review state and audit events; a revision returns to bilingual editing rather than restarting source acquisition. Pair-grouped split metadata isolates the held-out test branch from production indexes and QLoRA training.

3.3 *Retrieval and answer generation*

BM25 indexes Chinese and English question-answer fields and evidence. Dense search embeds a query with BGE-M3 and ranks chunks by vector similarity. Hybrid retrieval fuses keyword and dense rankings using reciprocal-rank fusion. The `approved_only` condition applies expert status as a database-level filter. Retrieved evidence is numbered and supplied to the local generator with an instruction to answer in the query language and cite evidence labels.

Following Cormack *et al.* (2009), the reciprocal-rank-fusion score for a candidate document d is defined in Equation (1):

$$\mathrm{RRF}(d) = \sum_{r \in \{\mathrm{BM25}, \mathrm{vector}\}} \frac{1}{k_0 + \mathrm{rank}_r(d)}. \tag{1}$$

In Equation (1), a document absent from a retriever's candidate list contributes zero for that retriever. The candidate pool is fixed at 50 per retriever and the conventional rank constant is `k0 = 60`; Equation (1) is applied before final top-k is varied. This separation ensures that a top-k ablation changes the amount of evidence delivered to the generator without changing the upstream search pool. Dense ranking uses BGE-M3 query embeddings and pgvector distance, while lexical ranking retains exact matches that are valuable

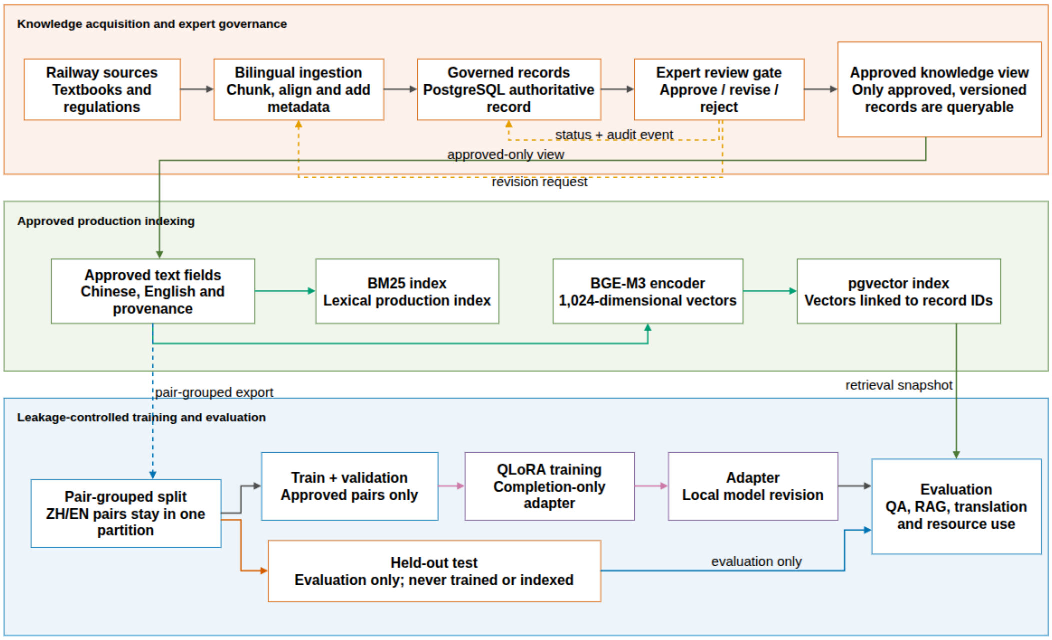

Figure 2: Expert-governed knowledge, production-index and leakage-controlled evaluation lifecycle. Source: Authors' own work.

for abbreviations, equipment names and regulation numbers. Every returned evidence object contains record identifier, source document, task type and retrieval score.

The FastAPI service exposes search and conversational endpoints to the Web client. At answer time, the selected evidence is serialised with labels such as `[Evidence1]` or `[证据 1]`. The system instruction requires the generator to answer in the query language, remain within the supplied evidence and attach labels to supported statements. If no admissible record is retrieved, the service returns an explicit no-evidence message rather than invoking unconstrained generation. Figure 1 maps learner, teacher and administrator workflows to the Web applications, online retrieval pipeline and local infrastructure. The output node is described as evidence-conditioned rather than inherently grounded because citation compliance and semantic support are evaluated separately.

3.4 Bilingual QLoRA adaptation

The adaptation dataset is generated only from approved records and split by knowledge-pair identifier, task type and source document. Chinese and English variants of the same knowledge item are assigned to the same partition. The current frozen 80/10/10 corpus contains 12,178 training, 1,524 validation and 1,526 test examples, balanced equally by language, with zero pair overlap among all partitions. The 120 regulation items reserved for RAG evaluation are forced into the test partition and never used for training. Loss is computed only on answer tokens; all prompt labels are masked with `-100`.

Qwen2.5-7B-Instruct and GLM-4-9B-Chat-HF form the adaptation matrix; their architectures and post-training lineage are documented in the corresponding official technical reports (Qwen Team, 2024; GLM Team, 2024). Both were loaded with NF4 quantisation and PEFT adapters on the target GPU. Qwen exposes separate `gate_proj` and `up_proj` modules, whereas GLM uses a fused `gate_up_proj`; model-specific target lists are therefore required. The formal rank-64 configuration trained 161,480,704 Qwen parameters and 190,382,080 GLM parameters, representing 3.58 and 3.45 per cent of the parameters visible to the quantised training process, respectively. Qwen3-14B is retained only as an unadapted reference generator and is associated with the Qwen3 technical report (Qwen Team, 2025).

Both runs use LoRA rank 64, alpha 16, dropout 0.05, learning rate `2e-4`, maximum sequence length 1,024 and one epoch. The per-device batch size is one with 16 gradient-accumulation steps and a 0.03 warm-up ratio. Random seed 42 is fixed in the experiment configuration. Generation uses greedy decoding (`temperature = 0`, `top_p = 1`) with a task-dependent output limit. Completion-only supervision masks every prompt token with `-100`; therefore, training optimises the answer sequence rather than teaching the model to reproduce instructions. This choice is tested rather than assumed to improve every downstream behaviour.

3.5 Evaluation design

Retrieval conditions are BM25, vector, hybrid and approved hybrid. Metrics are Recall@1/3/5, MRR and mean latency. The regulation pilot includes six answer conditions; the formal generator matrix fixes no retrieval, BM25-RAG and approved hybrid-RAG so that five generators receive identical cached contexts. Automatic metrics include answer F1, reference containment, citation-format coverage, a citation-format-based hallucination proxy and end-to-end latency. Citation measures are treated as instruction-following proxies, not evidence entailment. Translation is reported independently for Chinese-to-English and English-to-Chinese terminology and sentences using SacreBLEU, chrF++ and COMET (`Unbabel/wmt22-comet-da`). This multi-dimensional protocol follows the broader principle that LLM evaluation should separate task capability, robustness and deployment behaviour rather than rely on one aggregate score (Chang *et al.*, 2024).

Recall@k is one when the held-out source record occurs in the first k positions and zero otherwise; MRR averages the reciprocal rank of the first relevant record. Character-level F1 is used for Chinese and token-normalised overlap for English. Reference containment records whether the normalised reference appears in the generated answer. Citation coverage records only whether the requested evidence-label syntax appears. The hallucination proxy is zero when a recognised no-evidence response or citation label is present and one otherwise; it is deliberately named a proxy because it does not test claim-evidence entailment. This limitation is retained in every interpretation.

All core comparisons use paired samples. Mean metrics are accompanied by 2,000-resample bootstrap 95 per cent confidence intervals. Original-versus-QLoRA comparisons use two-sided paired Wilcoxon tests, paired standardised effect sizes and Holm correction across the pre-specified comparison family. Exploratory error categories are reported separately and are not treated as confirmatory hypothesis tests. Efficiency measurements use 30 Chinese short questions, 30 English short questions and 30 regulation-oriented long-context questions, with one warm-up and three measured repetitions per model condition.

Four automated system checks extend the primary protocol without introducing new human labels. First, source-only, Chinese-field, English-field and bilingual-field indexes are compared on the same 800 queries with a 512-token BGE-M3 window. Second, each of 8,000 RAG answers is segmented into claims; BGE-M3 cosine similarity estimates semantic support against all retrieved evidence and explicitly cited evidence at a pre-specified 0.45 threshold. This is a support proxy, not textual entailment or expert correctness. Third, immutable review events and before-state snapshots are audited for action and changed-field coverage. Fourth, the deployed API is warmed once per retrieval mode and tested with 12 retrieval-only requests at concurrency 1, 5 and 10. Client P50/P95, throughput, failures and GPU memory are recorded with speech synthesis disabled; generator efficiency remains covered by the separate 270-measurement inference experiment.

3.6 Implementation and reproducibility

The application uses a Vite-based Web client, a FastAPI backend and PostgreSQL 16.14 with pgvector 0.8.4. Experiments ran on Linux with an NVIDIA GeForce RTX 3090 (24,576 MiB), 32 GB host memory, Python 3.11.15, PyTorch 2.11.0, Transformers 5.12.1, PEFT 0.18.1 and bitsandbytes 0.49.2. Qwen2.5-7B-Instruct, GLM-4-9B-Chat-HF and BGE-M3 are pinned by model revision in the frozen manifest. COMET uses the pinned `Unbabel/wmt22-comet-da` checkpoint. Input datasets, configurations, generated tables and final analysis assets are recorded with SHA-256 hashes. Table I summarises the formal implementation and experimental environment.

Table I. Formal implementation and experimental environment.

Component	Formal setting
Database	PostgreSQL 16.14; pgvector 0.8.4
Embedding	BAAI/bge-m3; 1,024 dimensions
Adapted generators	Qwen2.5-7B-Instruct; GLM-4-9B-Chat-HF
Reference generator	Qwen3-14B through Ollama 0.19.0
Training	NF4 QLoRA; rank 64; one epoch; seed 42
Hardware	RTX 3090 24 GB; 32 GB RAM; single workstation
Statistical analysis	2,000 bootstrap resamples; paired Wilcoxon; Holm correction

The code separates test-set construction, embedding, retrieval evaluation, generator evaluation, adaptation and report export. This prevents an analysis script from silently changing the production index or training split. Cached retrieval contexts are reused across generators in the formal RAG matrix, so model comparisons receive identical evidence. The manifest records the Git state and every input hash at freeze time, supporting result reconstruction even when copyrighted source passages cannot be redistributed.

Production embedding loading is local-files-only by default, preventing an unavailable model registry from becoming a hidden runtime dependency. Audio synthesis remains enabled for normal users but is explicitly disabled in load tests. The real review and question-answering views were rendered through the running Vite and FastAPI services rather than reconstructed as mock-ups.

4 Results

4.1 Knowledge-base composition

The database contains 37,661 approved records and 37,109 complete bilingual records. After freezing the formal RAG test, 37,261 approved non-test knowledge chunks have BGE-M3 embeddings. The QLoRA test partition contains 763 knowledge pairs across four source documents. From this partition, the formal cross-source RAG set contains 400 knowledge pairs and 800 language-specific queries: 150 regulation, 127 terminology and 123 textbook/concept pairs across 17 task types. The original 120 regulation pairs remain a separately reported specialised subset.

4.2 Regulation-only pilot retrieval

In the 120-pair regulation-only pilot, approved hybrid retrieval produced Recall@5 of 0.758 in Chinese and 0.708 in English, compared with BM25 values of 0.742 and 0.667. English queries remained harder, supporting language-disaggregated reporting. These 120 knowledge pairs are a subset of the formal 400-pair set, so the pilot is retained only as a development-stage sensitivity check; it is not an independent sample, added as extra observations or used as the primary cross-source estimate.

4.3 Formal cross-source bilingual retrieval

Table II. Formal cross-source retrieval on 400 knowledge pairs.

Retrieval	Language	R@1	R@3	R@5	MRR	Mean latency (ms)
BM25	Chinese	0.520	0.595	0.620	0.560	**69.5**
Vector	Chinese	0.518	0.653	0.690	0.588	134.4
Hybrid approved	Chinese	**0.570**	**0.675**	**0.715**	**0.628**	217.6
BM25	English	**0.570**	**0.670**	0.685	0.620	**59.0**
Vector	English	0.463	0.553	0.580	0.509	134.4
Hybrid approved	English	**0.570**	0.668	**0.708**	**0.620**	209.2

Table II reports the formal cross-source retrieval comparison. The larger cross-source evaluation changes the interpretation of the pilot. BM25 is particularly competitive for English terminology queries, while dense retrieval contributes more strongly to Chinese evidence recall. With the RRF candidate pool fixed at 50 for every final cutoff, hybrid fusion obtains the highest Recall@5 in both languages, although its latency is approximately three times that of BM25. Increasing the final cutoff from three to eight raises approved-hybrid recall from 0.675 to 0.743 in Chinese and from 0.668 to 0.723 in English, while mean retrieval latency remains nearly constant because the candidate pool is controlled (Figure 3). Approved-only and unfiltered hybrid results are identical because all admissible indexed records in this frozen experiment satisfy the approval condition.

4.4 Regulation-only pilot answer generation

In the 120-pair regulation-only development pilot, no-retrieval F1 was 0.243 in both languages. The best pilot RAG F1 was 0.623 for Chinese vector-RAG and 0.498 for English BM25-RAG; citation-format coverage reached 0.950-0.992 across pilot retrieval conditions. RAG also reduced mean response time because evidence-constrained answers were shorter. Stronger hybrid evidence recall therefore did not automatically maximise answer F1, motivating the larger fixed-context generator experiment. Because the pilot pairs are already included in the formal 400-pair set, these values are reported as a configuration check and are not combined with or counted again in Table II or the formal generator comparisons.

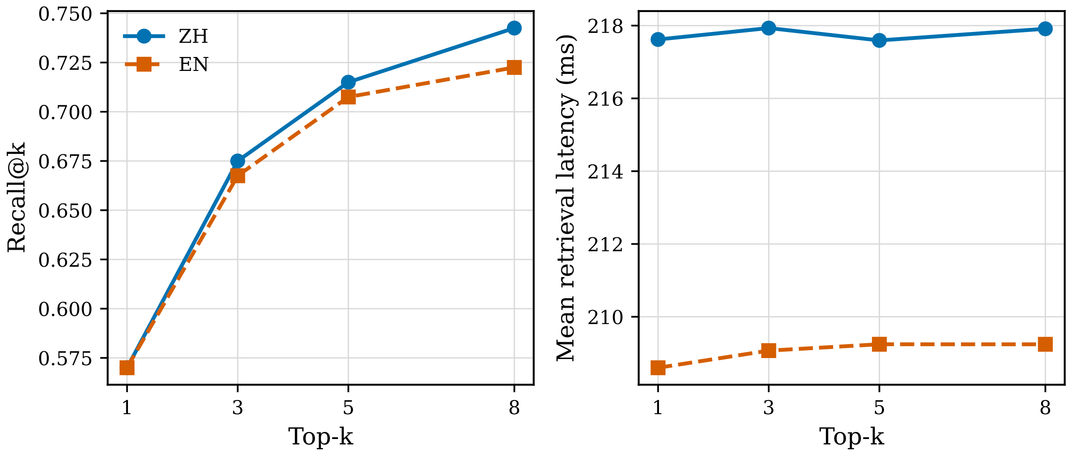

Figure 3: Approved-hybrid retrieval quality and latency across top-k settings. Left: Recall@k; right: mean retrieval latency. Source: Authors' own work.

4.5 QLoRA adaptation and held-out QA

Both one-epoch training runs converged without interruption. Qwen trained for 3,546 s (59.1 min), with final training and validation losses of 0.950 and 0.720 and peak reserved memory of 13.0 GB. GLM trained for 4,571 s (76.2 min), with losses of 1.146 and 0.794 and peak reserved memory of 15.0 GB. The respective validation perplexities were 2.05 and 2.21.

Figure 4 presents the training and validation trajectories. Because each model was trained for one epoch, the curves establish completion and optimisation behaviour rather than convergence across repeated schedules. The absence of an extended hyperparameter search is consistent with the resource-constrained design but limits claims about globally optimal adapter settings.

On the 1,526-example held-out bilingual QA set, Qwen QLoRA increased Char F1 from 0.189 (95 per cent CI 0.175-0.204) to 0.398 (0.376-0.422) in Chinese and from 0.437 (0.419-0.455) to 0.648 (0.632-0.664) in English. GLM increased from 0.192 to 0.410 in Chinese and from 0.498 to 0.552 in English. All four paired Char F1 gains remained significant after Holm correction (all adjusted p-values were below 0.01: 1.82e-40, 4.44e-48, 2.68e-44 and 3.76e-06 for Qwen Chinese, Qwen English, GLM Chinese and GLM English, respectively); paired effect sizes were 0.569 and 0.634 for Qwen Chinese and English, and 0.614 and 0.146 for GLM. Qwen therefore provides the strongest balanced adapted QA result, while the small GLM English effect cautions against pooling languages.

A limited general-capability check (maximum 200 examples per subtask) found that Qwen C-Eval/MMLU accuracy changed from 0.788/0.739 to 0.775/0.728, whereas GLM changed from 0.675/0.673 to 0.683/0.679. These small changes do not indicate broad catastrophic forgetting, but the limited protocol is a regression check rather than a comprehensive general benchmark.

4.6 Multi-generator RAG and interaction effects

Approved-hybrid RAG significantly improved Answer F1 over no retrieval for every generator and language after Holm correction. The adjusted p-values for Chinese and English, respectively, were 1.89e-39 and 2.28e-42 for original Qwen, 1.13e-22 and 9.56e-31 for original GLM, 1.34e-06 and 1.41e-11 for Qwen QLoRA, 3.16e-09 and 9.37e-03 for GLM QLoRA, and 6.66e-32 and 7.58e-29 for Qwen3-14B. For original Qwen, F1 rose from 0.262 to 0.402 in Chinese and from 0.306 to 0.440 in English (paired effect sizes 0.796 and 0.798). Qwen QLoRA achieved the highest absolute RAG F1, 0.660 in Chinese and 0.651 in English, exceeding the Qwen3-14B reference values of 0.442 and 0.454. However, its incremental hybrid-RAG gains over no retrieval were only 0.030 and 0.078, compared with 0.141 and 0.133 for original Qwen. Adaptation and retrieval are therefore complementary in answer overlap, but with diminishing returns.

The traceability result points in the opposite direction. Approved hybrid citation-format coverage was 0.595/0.643 for original Qwen and 0.788/0.905 for Qwen3 in Chinese/English, but 0.000/0.003 for Qwen QLoRA. GLM showed the same, less extreme pattern. QLoRA specialised answer content while weakening compliance with a citation instruction absent from its training targets. The citation-based hallucination proxy consequently approaches one for adapted models and must not be read as a factual hallucination rate. For an operational system, Qwen2.5 QLoRA is the primary answer-quality model, while original Qwen or Qwen3 remains the safer traceability control until citation-aware adaptation is added.

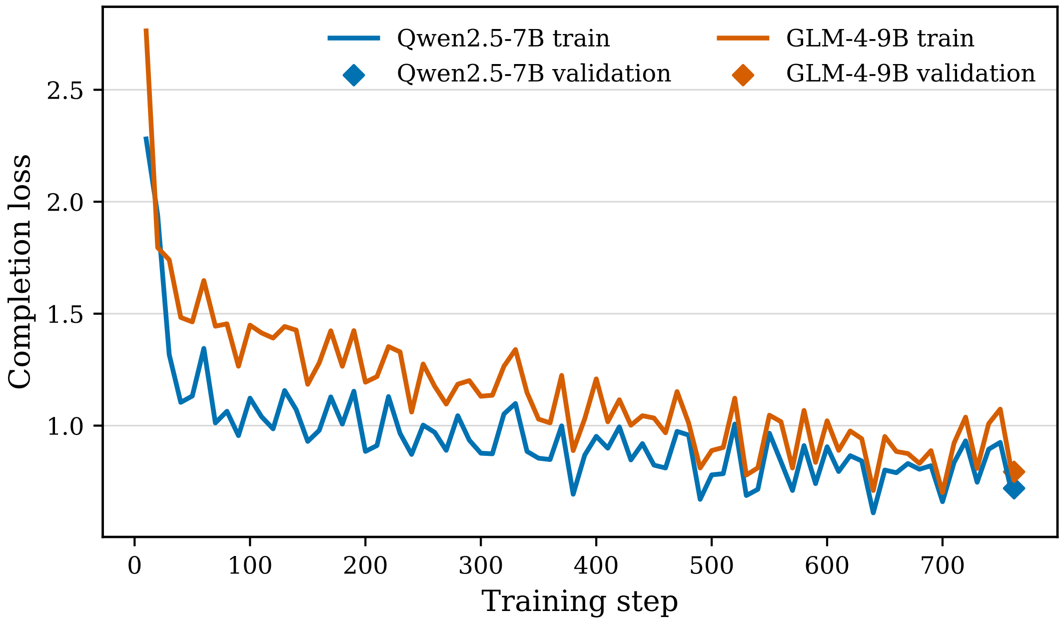

Figure 4: Completion-only QLoRA training and validation loss for Qwen2.5-7B and GLM-4-9B. Source: Authors' own work.

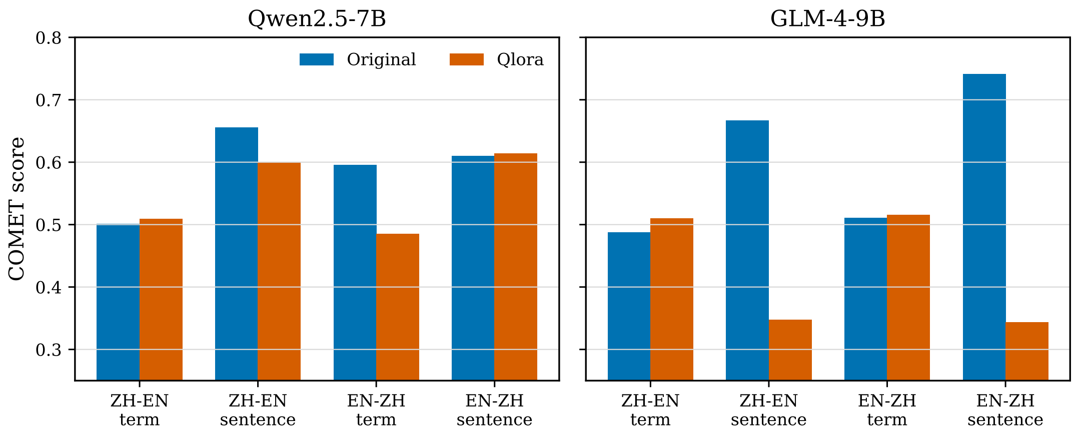

Figure 5: Direction- and task-separated COMET before and after QLoRA. Left: Qwen2.5-7B; right: GLM-4-9B. Source: Authors' own work.

4.7 Directional translation

Translation effects were direction- and task-dependent. Qwen COMET moved from 0.501 to 0.509 for Chinese-to-English terminology and from 0.610 to 0.614 for English-to-Chinese sentences, but fell from 0.656 to 0.599 for Chinese-to-English sentences and from 0.596 to 0.485 for English-to-Chinese terminology. Its lexical metrics show the same lack of uniform benefit: Chinese-to-English terminology chrF++ increased from 10.41 to 16.69, whereas sentence chrF++ fell from 47.43 to 31.28.

Figure 5 visualises the before-and-after COMET changes by direction and task. The opposing slopes demonstrate why the result cannot be summarised as a single improvement or regression.

GLM QLoRA is a clear failure case. Its sentence COMET dropped from 0.667 to 0.348 for Chinese-to-English and from 0.742 to 0.343 for English-to-Chinese; corpus BLEU was zero in both directions. Automated inspection found 2,168 empty outputs among 2,634 translation examples (82.3 per cent). This result rules out a claim that QA-oriented completion-only adaptation generally improves translation and shows why terminology and sentence translation must remain separate from QA evaluation.

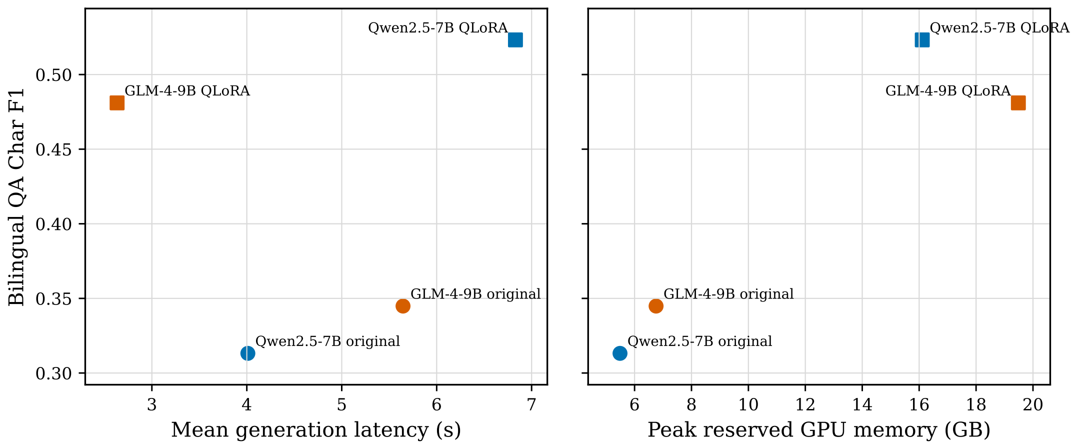

Figure 6: Bilingual QA quality against generation latency and peak GPU memory. Left: mean generation latency; right: peak reserved GPU memory. Source: Authors' own work.

4.8 *Resource use and automated error analysis*

All five deployment conditions completed 270 measurements without failure or OOM. Original Qwen and GLM used 5.49 and 6.76 GB peak reserved memory and generated at 31.9 and 22.7 tokens/s. Their QLoRA variants used 16.11 and 19.49 GB and generated at 17.5 and 11.9 tokens/s. Qwen3-14B through Ollama used 14.26 GB and 78.5 tokens/s, although cross-backend timing should be interpreted cautiously. The Qwen and GLM adapters occupy approximately 627 and 746 MB. The low 2.63 s GLM QLoRA mean latency reflects abnormally short or empty outputs and is not an efficiency advantage. All configurations fit the target 24 GB workstation, but original Qwen provides the lowest resource cost and Qwen QLoRA the strongest QA quality within the limit.

The quality-latency-memory relationship is shown in Figure 6. The plot is interpreted as a deployment frontier rather than a universal model ranking because the Transformer and Ollama inference implementations use different backends.

Automated failure flags explain several aggregate differences. At top three, approved hybrid retrieval missed the held-out item for 32.5 per cent of Chinese and 33.3 per cent of English queries. Even with a hit, original Qwen produced low-overlap answers in 18.0 per cent of Chinese versus 4.3 per cent of English cases. Qwen QLoRA reduced these rates to 4.0 and 1.5 per cent, but omitted citation labels in 100.0 and 99.8 per cent. Figure 7 compares these automatically flagged categories. The translation analysis additionally exposed empty answers, wrong-language outputs and terminology substitutions. Similar-version retrieval and citation entailment require expert semantic judgement and were not inferred from string-based flags.

4.9 *Index, evidence-support, governance and load validation*

Table III. Supplementary information-system validation results.

Validation	Chinese	English	Operational result
Bilingual-field hybrid index, Recall@5	0.718	0.675	Highest balanced mean (0.696)
Original Qwen hybrid, supported-claim proxy	0.878	0.836	Citation precision 0.955/0.970
Qwen QLoRA hybrid, supported-claim proxy	0.964	0.905	Citation recall 0.000/0.002
Governance history	-	-	1,337 events; 82 edits; two recorded reviewers

Validation	Chinese	English	Operational result
BM25 retrieval, concurrency 10	-	-	P95 0.515 s; 21.57 requests/s; 0/12 failures
Hybrid retrieval, concurrency 10	-	-	P95 1.125 s; 9.51 requests/s; 0/12 failures

Table III reports the supplementary information-system validations. The field ablation shows why both language fields are retained: English-only hybrid is strongest for English Recall@5 (0.713) but falls to 0.560 in Chinese, whereas the bilingual index gives the highest balanced mean. Across 27,668 segmented claims, hybrid retrieval generally increases automated semantic support, but high support does not repair absent citations: Qwen QLoRA retains strong support scores while almost never attaching a valid label. The governance database contains 37,664 records and 1,337 review events, including before-state snapshots for edits; only three current records are rejected, so no causal approved-versus-rejected quality claim is made. All 72 warmed retrieval requests succeed. BM25 offers the lower tail latency, while Hybrid approximately doubles throughput from concurrency 1 to 10 at the cost of higher P95 and 3.76 GB peak system GPU use. Figure 8 summarises these four checks.

5 Discussion

5.1 Answers to the research questions

RQ1 concerned bilingual retrieval. Lexical and semantic retrieval are language-dependent complements. Dense retrieval contributes most to Chinese recall, while exact terminology makes BM25 particularly competitive in English. Approved hybrid fusion reaches the highest Recall@5 in both languages, and the field ablation shows that a bilingual index offers the strongest balanced result even though an English-only index is best for English alone. Hybrid's higher latency means lexical retrieval remains a credible low-cost configuration. This result supports combining retrieval signals and language fields rather than assuming semantic embeddings dominate exact technical language (Robertson and Zaragoza, 2009; Cormack *et al.*, 2009).

RQ2 concerned retrieval strategy, governance and answer behaviour. RAG improves answer overlap for every generator, but stronger retrieval recall does not always yield the highest F1. QLoRA and RAG are complementary for answer content but not automatically for traceability: adaptation reduces the marginal F1 gain from retrieval and can overwrite citation-format behaviour. Automated semantic support remains high for adapted answers while citation recall approaches zero, demonstrating that relevance to retrieved evidence and visible attribution are separate requirements. This qualifies railway domain systems that report aggregate accuracy gains from fine-tuning or RAG (Li *et al.*, 2024; Luo *et al.*, 2025; Huang *et al.*, 2026). Citation-aware targets, constrained citation insertion and human-validated entailment checking are required before operational use.

The formal generator matrix confirms that approved-hybrid RAG improves Answer F1 over no retrieval for every evaluated generator in both Chinese and English, while the magnitude of the gain remains model- and language-dependent.

The approved-only and unfiltered hybrid conditions are identical because every admissible record in the frozen production index was already approved. This confirms enforcement of the governance boundary but does not estimate the causal quality benefit of review. Such an estimate would require a deliberately retained, ethically usable unreviewed comparison corpus. The contribution here is therefore the executable review-state mechanism and its auditability, not a causal claim that approval alone raises F1.

RQ3 concerned bilingual adaptation and translation. Completion-only QLoRA substantially improves held-out QA, particularly for Qwen, but adaptation quality is task-specific. QA gains coexist with asymmetric Qwen translation regressions and severe GLM sentence-translation failure. This mirrors concerns that domain-limited fine-tuning can overfit task form or translation direction (Hu *et al.*, 2024; Vieira *et al.*, 2024). The practical implication is that QA, terminology translation, sentence translation and citation compliance need independent acceptance thresholds; no pooled bilingual score can justify deployment.

RQ4 concerned constrained deployment. Every generator condition fits a 24 GB workstation, and the deployed retrieval API completes 72 of 72 steady-state requests at up to ten concurrent clients. BM25 sustains about 21.6 requests/s with 0.515 s P95 at concurrency ten; Hybrid reaches 9.5 requests/s with 1.125 s P95. Nevertheless, an adapter's small file size does not imply lower runtime memory: both QLoRA variants reserve more memory and generate more slowly than their quantised base models. Resource reporting must

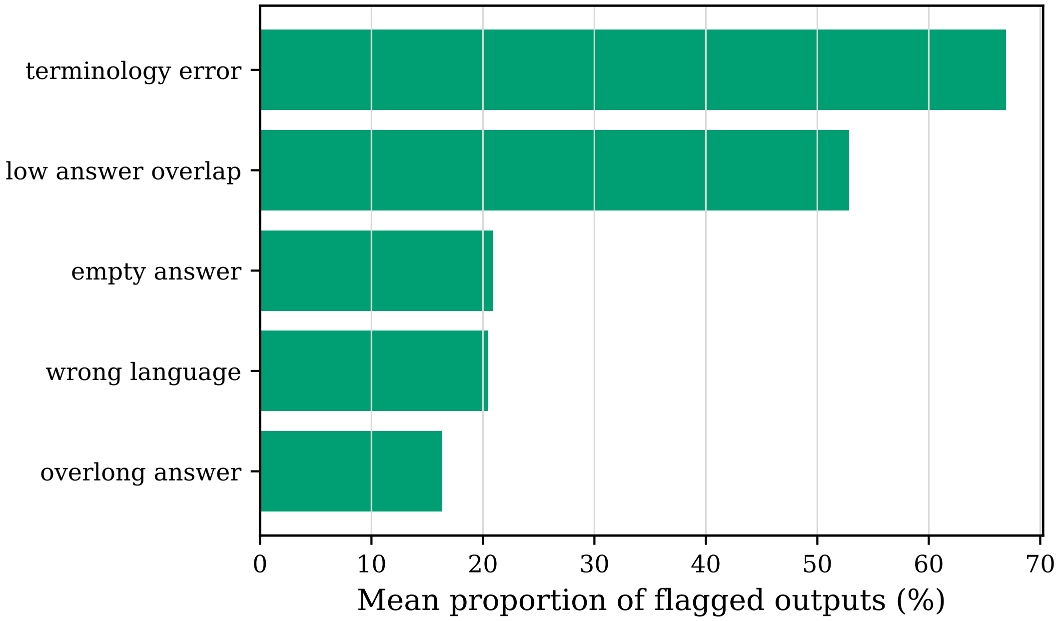

Figure 7: Mean prevalence of automatically flagged output errors across the evaluated generator and retrieval conditions. Source: Authors' own work.

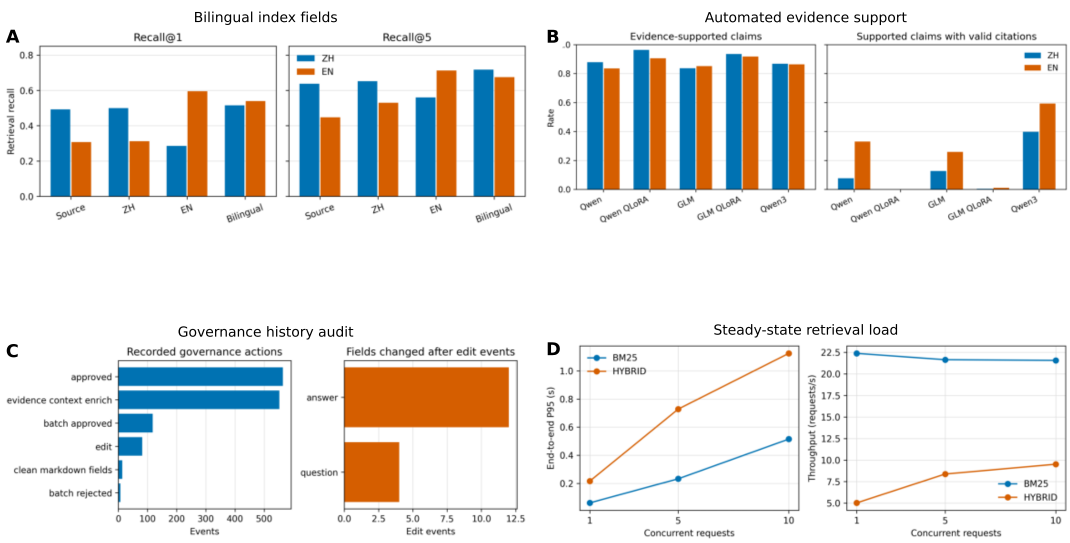

Figure 8: Bilingual index, evidence support, governance history and steady-state retrieval load validation. Panels A-D report field ablation, evidence support, governance history and load results, respectively. Source: Authors' own work.

therefore include measured inference and service behaviour, not only trainable parameter counts (Ding *et al.*, 2023; Lv *et al.*, 2024).

5.2 Implications for Web information-system design

The principal theoretical implication is that domain knowledge enhancement should be evaluated as an information-system pipeline rather than a single-model treatment. Record governance determines admissible evidence; retrieval determines whether that evidence is present; generation determines whether it is used; and the interface determines whether provenance is visible. Reporting only final answer overlap collapses these distinct failure modes. The layered protocol used here reveals, for example, that hybrid retrieval can improve recall without maximising answer F1, and that adaptation can improve F1 while nearly eliminating citation-format compliance.

The architecture also treats relational and vector data as complementary. PostgreSQL remains the authoritative store for source identity, review state and test exclusion, while pgvector adds semantic access without replacing those controls. This arrangement is suitable for institutional Web information systems whose regulations, teaching materials and language fields evolve over time. It permits a record to be revised or withdrawn through ordinary data governance while preserving reproducible model and embedding versions.

6 Practical implications

The architecture can support institutions that need local control of teaching resources and cannot rely on an external hosted model. PostgreSQL review status, source metadata and held-out flags make governance rules executable at retrieval time. On a 24 GB workstation, original Qwen is the practical low-cost option, Qwen QLoRA is preferred when answer overlap is the priority, and a citation-compliant original model should remain in the workflow when source display is mandatory. Teachers should inspect evidence and citations rather than treating either F1 or citation presence as correctness. International-student interfaces should retain both source language and translated fields because retrieval and translation errors are direction-specific.

Deployment can be staged without retraining the entire system. In the first stage, institutions can expose reviewed BM25 retrieval and bilingual source display, which offers the lowest latency and preserves direct evidence access. In the second stage, BGE-M3 and hybrid fusion can be enabled for questions whose wording differs from source terminology. In the third stage, local generation can be added with explicit no-evidence behaviour and teacher-visible citations. Adapted models should enter service only after passing separate QA, citation and translation gates. This staged path reduces the risk of treating generative output as the system of record.

Role-specific interfaces are also important. International learners need language selection, concise answers and side-by-side source evidence. Teachers need record editing, source-page inspection, review status and revision history. Administrators need index coverage, model revision, latency, memory and failed-generation monitoring. Because the underlying record identifier is preserved across these views, feedback can be attached to the responsible knowledge item rather than disappearing into an untraceable chat log.

For railway education, the system should be framed as decision and learning support rather than an authority for live safety-critical action. A visible source citation allows a teacher or learner to consult the controlling regulation, but neither a citation label nor an automatic F1 score establishes current legal or operational validity. Version dates, withdrawal state and responsible reviewer should therefore remain visible when the knowledge base is updated.

7 Limitations

The held-out set covers four source documents and 17 task types, but it remains specific to one railway vocational knowledge base and cannot represent every course, regulation version or learner need. The database approval field is an operational governance control; the present author-led review does not establish independent inter-rater agreement or claim that every approved answer reaches external expert consensus. Because the frozen index contains only approved admissible records, the experiment cannot quantify an approved-versus-unreviewed quality effect. Automatic F1 penalises valid paraphrases, while citation presence and the hallucination proxy do not establish entailment. BGE-M3 claim-evidence similarity is an automated semantic support proxy and its 0.45 threshold has not been calibrated against expert judgements. The error analysis is rule-based and no new independent human scoring was conducted. General-benchmark checks were capped per subtask.

The study also does not measure learner outcomes, usability, generation throughput under concurrent users or long-term knowledge-maintenance cost. The load test isolates warmed retrieval with speech synthesis disabled and therefore does not represent wide-area networking or simultaneous local generation. The five generators do not represent all current multilingual models, and the one-epoch rank-64 setting does not establish an optimal PEFT configuration. Some source passages cannot be redistributed because of third-party rights, limiting full public replication of the corpus even though split hashes, scripts and aggregate results can be released. Finally, cross-backend speed comparisons between the Transformer and Ollama inference implementations are descriptive rather than controlled systems benchmarks.

8 Conclusion

This study demonstrates a bilingual railway education Web information system that combines governed PostgreSQL records, pgvector/BGE-M3 retrieval and local generation. Hybrid retrieval achieved the strongest final recall in both languages, and bilingual fields produced the strongest balanced index ablation. Qwen2.5 QLoRA produced the strongest held-out QA and RAG answer-overlap scores within a 24 GB GPU limit, while the warmed retrieval service completed all requests through ten-client concurrency. The same adaptation weakened citation compliance and did not consistently improve translation; GLM QLoRA failed substantially on sentence translation. The principal design implication is therefore not to choose between adaptation and RAG, but to govern them as separate, testable components with language-, task-, load- and traceability-specific acceptance criteria.

The contribution to Web information systems lies in making those acceptance boundaries executable: reviewed records, held-out exclusions, embedding revisions and evidence provenance are represented in the data and retrieval layers rather than left to a language-model prompt. Future work should complete the documented expert-panel metadata, calibrate citation entailment with human judgements, conduct learner-centred evaluation and test concurrent generation across additional railway curricula and language pairs.

9 Declarations

Data availability: Evaluation scripts, experiment configurations, aggregate metrics and non-copyrighted derived test identifiers can be requested from the authors. Full source passages from textbooks and regulations cannot be redistributed where third-party copyright or licensing restrictions apply. Researchers seeking restricted evaluation materials or additional reproducibility information may contact the authors through the journal submission system.

Ethics and consent: No learner personal data are used in the current experiments.

Conflict of interest: The authors declare no conflict of interest.

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Use of generative AI: Generative AI tools were used solely to copy-edit the authors' pre-existing original manuscript for language, grammar and structural clarity. No new scholarly content, research design, data, analysis, results, references or conclusions were generated by AI. The authors reviewed and approved every change and take full responsibility for the accuracy, originality and integrity of the manuscript.

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